

Dried tomatoes

Sun drying is used in many parts of the world for preserving certain foods, such as fruits and nuts. However, this method can be used only if the *climatic conditions are hot* with low humidity. In many cases foods are pretreated before drying to make the structure more porous and to facilitate transfer of moisture, thereby speeding the drying rate. Food porosity increases the chance of quick solubility on reconstitution, but is at a disadvantage due to increased bulk and shorter storage stability. *Vegetables like beans, peas, potatoes, cauliflower, ladies finger, garlic, onion and all leafy vegetables can be sundried.*

Changes during drying

- Shrinkage occurs on the surface first and then proceeds to the inner layers. With quick high temperature drying of food, the surface becomes dry and rigid long before the center dries out.
- Dried food pieces may also contain cracks and pores of various diameters. The shrinking and pore clogging by the solutes is known as core hardening. It can be minimized by gradual drying with low surface temperature.
- Foods that lack good structure and are high in sugar content, give an impression of retaining moisture even after the drying process. Fruits like grapes and figs have high sugar content and lack good structure, hence appearing moist even after dehydration.
- Complete prevention of these changes is impossible. They can be minimized by using appropriate technology.

Dry figs are richer in fiber. A ¼ – cup serving of dried figs is nutritionally comparable to a serving of two large fresh figs, providing about same amount of calories, fibres and potassium.

Methods of Drying

A number of drying methods are available; some are suitable for liquids, others for solid foods or mixtures containing food pieces. The common drier types used for liquid and solid foods may be categorized as the air-convection drier, drum or roller drier and vacuum drier.

4.2.3.2 Types of Driers

- **Air-Convection drier** – In the air-convection drier, hot air supplies the heat for evaporation. Though there are different types of air-convection driers, they all have an insulated enclosure, a means of circulating air through the enclosure and a means of heating this air.

  If liquid, the food may be sprayed or poured into pans or on belts. Food in the form of a fine spray or mist is

Chapter 4_070-087.indd 78 31-03-2018 13:21:40